TEXORA · SESSIONAL STUDY SHEET

Pipe Fittings, Steam Generating Unit, Pumps, Compressor, Power Generator, Engines and Turbines

One combined study of the mill's utility hardware: identify each fitting and machine, state what it does, what to observe and record on the plant visit, and how the pieces join into the steam-and-power chain from boiler house to driven shaft.

ME-3204 · THERMAL ENGINEERING AND FLUID MECHANICS SESSIONAL · STUDY 1 · LEVEL 3 TERM 2

1. How to work this study

For every item below the routine is the same: **identify** the part on the actual plant (photograph or sketch it), **name its function** in one sentence, and **record the duty data** from its nameplate or gauge board — pressure, temperature, capacity, speed, fuel. Utility plant is learned standing in front of it with a notebook; the tables below are the checklists to carry. Safety rules first, always: never open a fitting on a live steam line, treat every unlagged surface as hot, and take gauge readings from behind the gauge glass line, not over it.

2. Pipe fittings and valves

FITTING / VALVE	FUNCTION — WHAT TO IDENTIFY AND RECORD
Elbow (90°/45°), tee, cross, reducer	Change direction, branch, join different sizes. Note screwed vs flanged vs welded joints and pipe material (GI, MS, SS)
Union, coupling, flange pair	Demountable joints — unions for small screwed lines, flanges (with gasket) where equipment must be removable
Gate valve	Full-open/full-shut isolation; straight-through flow, low loss when open. Never throttle with it. Identify by rising stem and wedge body
Globe valve	Throttling and flow regulation — tortuous S-path gives control at the price of pressure drop. Steam lines and dosing control
Non-return (check) valve	Passes flow one way only; protects pumps and boilers from backflow. Swing and lift patterns; note the arrow on the body

Ball / butterfly valve	Quarter-turn quick shut-off; ball for small lines, butterfly for large air and water ducts
Safety (relief) valve	Spring-loaded guardian: lifts at set pressure, reseats below it. Record set pressure — on a boiler it is law, not adjustment
Steam trap	Discharges condensate and air but not live steam (float, thermodynamic, thermostatic patterns) — the invisible fuel-saver of every steam main
Expansion loop / bellows	Absorbs thermal growth of steam lines; a 30 m steel main grows ≈ 55 mm between cold and working temperature — something must give
Strainer (Y-type)	Mesh basket protecting traps, valves and pumps from scale and debris; the first thing to check when a line underperforms

3. The steam generating unit (boiler)

Identify the boiler type first. **Fire-tube** designs (Lancashire, Cochran, packaged three-pass) run hot gas through tubes surrounded by water — compact, forgiving, the standard mill boiler up to moderate pressures. **Water-tube** designs run water through tubes inside the furnace — higher pressures and capacities, faster steam raising, the utility choice. Then walk the fittings, dividing them as the boiler code does:

- **Mountings** (fitted to the shell, safety-critical): safety valves (two), main steam stop valve, feed check valve, water gauge glass with protector, pressure gauge with siphon, blow-down valve, fusible plug (fire-tube).
- **Accessories** (efficiency equipment): economiser (flue gas preheats feedwater), air preheater, superheater (raises steam temperature for turbines), feed pump and injector.
- **Record:** rated pressure and evaporation (kg/h), fuel (gas/oil/coal/biomass husk), and the efficiency estimate $\eta = m_s(h_s - h_{feed})/(\text{fuel rate} \times CV)$ if plant data allows.

> **WORKED CHECK — BOILER EFFICIENCY FROM PLANT DATA**
>
> A gas-fired boiler evaporates 2000 kg/h of steam at h = 2796 kJ/kg from feedwater at h = 419 kJ/kg, burning 160 m³/h of gas of calorific value 38,000 kJ/m³. Heat to steam = 2000 × (2796 − 419) = 4.754×10^6 kJ/h. Heat in fuel = 160 × 38,000 = 6.08×10^6 kJ/h. η = 4.754/6.08 = **78.2%** — a believable figure for a maintained packaged boiler, and the calculation is the template for the study report.

4. Pumps and compressor — the movers

On the pump set (theory in ME-3203 Topic 4), identify: centrifugal pattern (end-suction vs multistage), priming arrangement, suction and delivery gauges, and nameplate Q, H, rpm, kW. Watch a start: delivery valve cracked open, casing vented, prime confirmed. For the **compressor**, identify reciprocating vs rotary (screw/vane): the reciprocating machine compresses in a cylinder with automatic suction/delivery valves, usually in stages with intercooling for high pressure (instrument air at ~7 bar); rotary screw machines dominate modern plant air. Record receiver pressure, unload/load set points, and find the air dryer and oil separator in the line — wet, oily air wrecks pneumatic controls and marks fabric.

5. Engines, power generator and turbines — prime movers

MACHINE	IDENTIFYING FEATURES & WHAT TO RECORD
Petrol engine (Otto cycle)	Spark ignition, carburettor/throttle body, low compression ratio (~8–11). Small sets, vehicles. Four-stroke sequence: suction–compression–power–exhaust, one power stroke per two revolutions per cylinder
Diesel engine	Compression ignition — no spark plugs; injector pump/rail, glow plugs, high compression (~14–22), turbocharger on larger sets. The mill’s standby workhorse
Gas engine	Spark-ignited but fuelled by natural gas — common in Bangladeshi mill captive power; record gas pressure train and synchronising panel
Power generator (alternator)	Coupled to the engine/turbine: converts shaft work to AC. Record kVA rating, voltage (400 V), frequency (50 Hz), excitation type. Diesel-generator (DG) set = engine + alternator on one bedplate
Steam turbine	Impulse (pressure drops in nozzles — de Laval/Rateau) vs reaction (pressure drops in moving blades too — Parsons) stages; identify governor, steam chest, condenser (surface type) and gland sealing. Drives alternators or large machines
Gas / hydro turbines (awareness)	Gas turbine: compressor–combustor–turbine on one shaft (open cycle) in captive plants. Hydro: Pelton (high head), Francis (medium), Kaplan (low head) in national generation

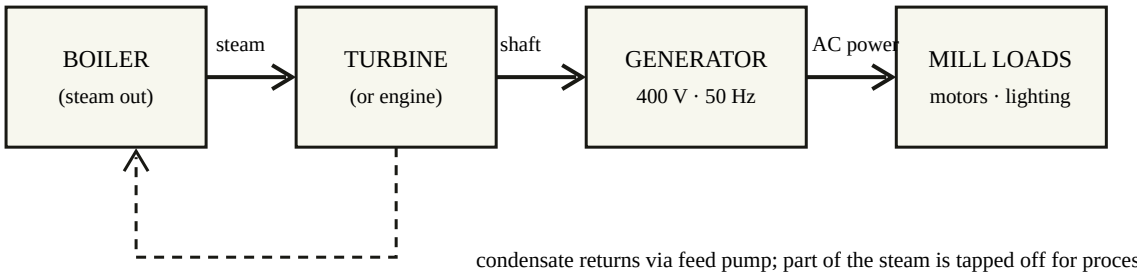


Figure 1 — The utility chain studied in this sheet: boiler → prime mover → generator → mill loads, with the condensate loop closing back to the boiler.

6. Key points

- Fittings: gate valves isolate, globe valves throttle, check valves protect, safety valves guard, traps drain condensate without wasting steam.
- Boilers: fire-tube for mill duty, water-tube for high pressure/capacity; mountings are safety devices, accessories are efficiency devices; plant-data efficiency ≈ 78% in the worked check.
- Compressors: reciprocating for high pressure in stages, screw for plant air volume; dry, clean air protects pneumatics and fabric.
- Prime movers: petrol/diesel/gas engines by ignition and fuel; impulse vs reaction steam turbines; the alternator’s nameplate (kVA, 400 V, 50 Hz) closes the chain.
- Every study entry ends in recorded numbers — pressure, temperature, capacity, set point — not adjectives.

7. Practice

Answer in your own words; brief model answers follow.

1. A line must be isolated completely for maintenance and a second line needs its flow trimmed daily. Choose valve types for each and justify.
2. Steam leaks visibly from a trap’s vent whenever the main is live. What has failed, and what does it cost?
3. List the boiler mountings that protect against over-pressure and against low water, and what each does.
4. How do you distinguish a diesel engine from a petrol engine standing in front of both, without starting either?

5. A DG set is rated 500 kVA at 0.8 power factor. What real power (kW) can it deliver, and why are both numbers quoted?
6. Why is a standby steam line fitted with an expansion loop rather than being anchored rigidly at both ends?

Answer key

1. Isolation: gate valve (full bore when open, positive shut-off, minimal loss). Trimming: globe valve (its S-path disc/seat is designed for throttling duty; a gate valve throttled part-open erodes its seat and vibrates).
2. The trap has failed open (or its seat is fouled): live steam blows straight to the condensate line. Cost is fuel — a single blowing trap can waste several percent of a boiler's output continuously, besides over-pressuring the return line.
3. Over-pressure: the safety valves (spring-loaded, lift at set pressure) — plus the pressure gauge that warns the operator. Low water: the gauge glass (shows the level), fusible plug (melts and vents the furnace if the crown is uncovered), and low-water alarm/cut-out where fitted.
4. Diesel: injector pump or common rail and injectors on the head, glow plugs, no spark-plug leads or distributor/coil pack, heavier construction, higher compression hardware (and usually a turbocharger). Petrol: spark plugs with HT leads, throttle body/carburettor, ignition coil.
5. Real power = 500 × 0.8 = 400 kW. Alternators are thermally limited by current (kVA) while engines are limited by real power (kW); loads with poor power factor draw more current per kW, so both ratings must be respected.
6. Steel expands ≈ 12×10^{-6} per °C; a long main heated through ~180 °C grows centimetres. Rigid anchors convert that growth into enormous compressive stress — buckled pipe, torn flanges. The loop flexes and absorbs the movement elastically.
